



ECE 3200: Electronics Project Design/Development

Smart Solar Tracker: A Multi-Functional Renewable Energy System

Name: Rukaiya Alam Nidra

Roll: 2109019

Name: Debraj Kundu

Roll: 2109052

Bachelor of Science in Electronics and Communication Engineering

Department of Electronics and Communication Engineering

27 April, 2026

Declaration

I hereby declare that this project report is my own work and has not been submitted elsewhere for any academic purpose.

Supervisor Name: _____

Designation: _____

Signature: _____

Student Name: _____

Roll: _____

Date: _____

Student Name: _____

Roll: _____

Date: _____

Acknowledgement

I would like to express my sincere gratitude to my supervisor and faculty members for their guidance and support throughout this project. I also thank my peers and family for their encouragement.

Abstract

This project presents the design and implementation of an Arduino-based solar tracking system with energy storage and multi-output applications. The system uses LDR sensors and a microcontroller to automatically align the solar panel toward maximum sunlight, improving energy efficiency compared to fixed panels.

The generated energy is managed using a TP4056 charging module to safely charge a lithium-ion battery. The stored energy is then utilized for mobile charging through a boost converter and for operating a water pump using a combination of boost and buck converters.

Experimental results show improved energy harvesting, stable battery charging, and reliable performance of both output applications. The system provides a cost-effective and eco-friendly solution suitable for off-grid and small-scale agricultural use.

Chapter 1: Introduction

The increasing global demand for energy and the depletion of conventional fossil fuels have led to a significant shift toward renewable energy sources. Among these, solar energy is one of the most abundant, clean, and sustainable sources available. However, the efficiency of traditional solar systems is often limited due to their fixed orientation, which prevents them from continuously facing the sun.

To address this limitation, solar tracking systems are employed. These systems dynamically adjust the position of solar panels to maintain optimal alignment with the sun throughout the day, thereby maximizing energy generation. The integration of microcontroller-based automation, such as Arduino, further enhances system efficiency and flexibility.

In this project, an Arduino-based solar tracking system is developed that not only improves energy harvesting but also incorporates energy storage and multi-output utilization. The generated energy is stored in a rechargeable battery using a dedicated charge controller and is utilized for two primary applications:

- Mobile device charging
- Water pumping for plantation or irrigation systems

The system integrates various components such as LDR sensors for light detection, motor control for tracking, TP4056 charge controller for battery management, and boost-buck converters for voltage regulation. This makes the system a complete standalone renewable energy solution, particularly suitable for off-grid and rural applications.

Chapter 2: Circuit and Methodology

The system is divided into three main sections:

- (i) Tracking System
- (ii) Energy Storage System
- (iii) Output Utilization System

2.1 Tracking System Circuit

- Two LDRs are connected in a voltage divider configuration
- Output of LDR circuits → Arduino analog pins (A0, A1)
- Arduino processes light intensity difference
- A servo motor is connected to Arduino PWM pin

Working:

- Unequal light intensity → Arduino generates control signal
- Motor rotates panel toward higher light intensity

2.2 Energy Storage Circuit

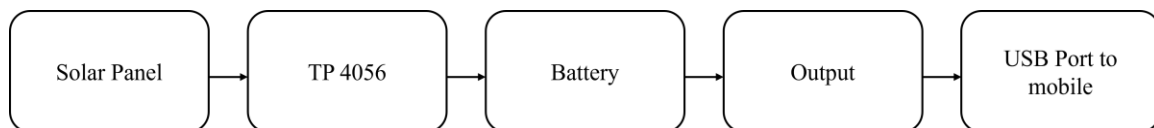
- Solar Panel output → TP4056 module (IN+ and IN-)
- TP4056 → Lithium-ion Battery (BAT+ and BAT-)

Features:

- Overcharge protection
- Over-discharge protection
- Stable charging current

2.3 Output Utilization Circuit

► Mobile Charging Path:



► Water Pumping Path:

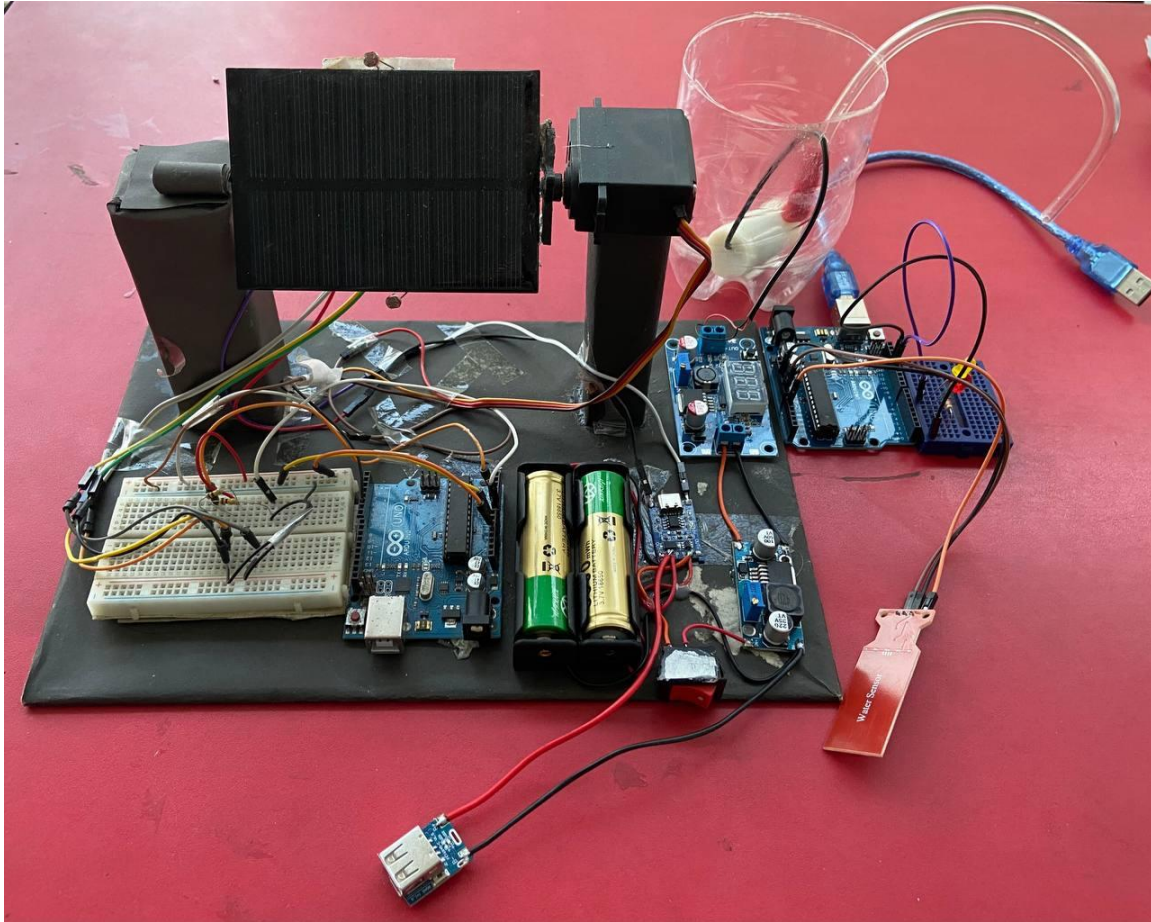
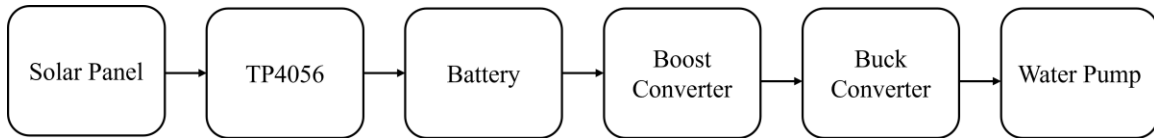


Fig 1 : Smart solar system with water pumping solution

Methodology

The proposed solar tracking-based energy system is developed through a structured sequence of hardware and control processes to ensure maximum solar energy harvesting, efficient storage, and practical utilization. The complete operation is divided into several interconnected stages as described below:

Step 1: Solar Energy Collection (Photovoltaic Conversion):

The system begins with the solar panel, which acts as the primary energy harvesting unit. When sunlight falls on the photovoltaic (PV) cells, it is converted directly into electrical energy through the photovoltaic effect. The amount of generated power depends on the intensity and angle of incident sunlight. To maximize output, the system aims to maintain the panel perpendicular to the sun throughout the day.

Step 2: Light Detection and Direction Sensing:

To track the sun's position, multiple Light Dependent Resistor (LDR) sensors are placed at different orientations around the solar panel. Each LDR continuously measures the intensity of incoming light. Since sunlight intensity varies depending on direction, the system compares the analog signals from all sensors. The Arduino microcontroller processes these values and identifies the direction where light intensity is maximum, which corresponds to the sun's position.

Step 3: Intelligent Panel Alignment (Tracking Mechanism):

Based on the processed LDR data, the Arduino generates control signals that are sent to a motor driver circuit. This driver activates a DC motor or servo motor connected to the solar panel structure. The motor rotates or tilts the panel in small incremental steps toward the direction of highest light intensity. This closed-loop feedback system ensures continuous alignment with the sun, improving energy harvesting efficiency significantly compared to fixed panels.

Step 4: Power Conditioning and Battery Charging:

The electrical output from the solar panel is not directly suitable for battery storage due to voltage fluctuations. Therefore, the output is fed into a TP4056 charging module. This module regulates the charging current and voltage, ensuring safe and stable charging of a lithium-ion battery. It also protects the battery from overcharging, over-discharging, and short-circuit conditions.

Step 5: Energy Storage System:

The lithium-ion battery serves as the energy storage unit of the system. It stores the excess electrical energy generated during peak sunlight hours. This stored energy is crucial for continuous power supply during nighttime, cloudy weather, or low sunlight conditions, making the system reliable and independent of direct sunlight availability.

Step 6: Power Utilization and Load Distribution:

The stored energy is utilized to power different practical applications. In this system, two main loads are considered:

- Mobile charging unit for portable electronic devices
- Water pumping system for agricultural or domestic use

The energy is distributed according to load demand, ensuring efficient utilization of stored power.

Step 7: Voltage Regulation and Conversion:

Since different loads require different voltage levels, voltage regulation circuits are integrated into the system:

- A boost converter is used when higher voltage is required than the battery output.
- A buck converter is used to step down voltage for sensitive electronic devices.

These converters ensure stable and safe operation of all connected loads.

Step 8: System Integration and Automation:

Finally, all subsystems—solar tracking, energy conversion, battery charging, storage, and load distribution—are integrated into a single automated unit. The Arduino acts as the central controller, coordinating all operations in real time. This integration ensures a fully automated, efficient, and sustainable renewable energy system with minimal human intervention.

Chapter 3: Budget Analysis

Name of Components	Quantity	Rating	Price(tk)
Solar Panel	1	6V,2W	225
Arduino	1		300
TP4056	1		50
Lithium ion battery	2	3000maH	200

Boost Converter	1		150
Buck Converter	1		230
USB Module	1		50
Water Pumping Motor	1		65
Servo Motor	1		200
Connecting Wire	20	-	200
			Total=

Chapter 4: Result and Discussion

4.1 Tracking Efficiency:

The implemented solar tracking system demonstrated effective and reliable performance in following the sun's movement throughout the day. The LDR-based sensing mechanism successfully detected variations in light intensity from different directions, allowing the Arduino-controlled motor system to continuously adjust the panel orientation toward the position of maximum sunlight exposure.

When compared with a fixed (stationary) solar panel, the tracking system showed a significant increase in energy output, with an estimated efficiency improvement of approximately 30–45% depending on weather conditions and time of day. The improvement was most noticeable during morning and evening hours when the sun's angle changes rapidly.

This clearly confirms that the automatic tracking mechanism enhances solar energy harvesting efficiency and provides a practical advantage over conventional fixed solar installations.

4.2 Battery Charging Performance:

The battery charging subsystem performed consistently under varying sunlight conditions. The TP4056 charging module regulated the charging process effectively by maintaining stable voltage and current levels suitable for lithium-ion batteries.

During operation:

- The battery charged smoothly under direct sunlight conditions.

- No overheating or overcharging issues were observed due to built-in protection features.
- The charging rate varied depending on solar intensity, with higher charging efficiency during peak sunlight hours.

Overall, the battery management system ensured safe, stable, and reliable energy storage, making it suitable for long-term use in off-grid applications.

4.3 Voltage Regulation Performance:

The voltage regulation stage played a crucial role in maintaining stable power output for different applications. The system incorporated both boost and buck converters to manage voltage levels according to load requirements.

- The boost converter successfully increased the voltage when the solar output or battery voltage was insufficient for certain applications.
- The buck converter ensured stable and reduced voltage output, particularly for sensitive electronic devices requiring regulated supply.

Throughout operation, the output voltage showed minimal fluctuations even under varying load conditions.

This confirms that the voltage regulation system is effective in ensuring safe and consistent power delivery across all connected components.

4.4 Mobile Charging Performance:

The mobile charging unit functioned successfully as a practical application of the system. The regulated output maintained a stable voltage close to **5V**, which is suitable for most mobile devices.

Observations include:

- Mobile phones charged without interruption.
- Charging speed was moderate, depending on battery level and solar input.
- The system maintained consistent output without voltage spikes or drops.

This demonstrates that the system is well-suited for off-grid mobile charging applications, especially in rural or remote areas.

4.5 Water Pump Operation:

The water pumping subsystem effectively demonstrated the system's capability for small-scale agricultural applications. When sufficient energy was available, the pump operated smoothly and delivered a consistent water flow.

Observations:

- Pump performance was stable under fully charged battery conditions.
- Water flow rate remained consistent during normal operation.
- Performance gradually decreased when battery charge level dropped.

This indicates that the system is suitable for low-power irrigation or domestic water supply applications but requires sufficient stored energy for continuous operation.

4.6 Discussion

The overall performance of the system validates the effectiveness of integrating solar tracking with energy storage and regulated power distribution. The following key conclusions can be drawn:

- **Enhanced Energy Harvesting:** Solar tracking significantly improves the amount of harvested energy compared to fixed systems.
- **Efficient Energy Management:** The combination of TP4056 battery charging and voltage regulation ensures safe and optimized energy utilization.
- **Multi-Application Capability:** The system supports multiple outputs such as mobile charging and water pumping, increasing its practical usefulness.

However, the system also has certain limitations:

- Performance is highly dependent on availability of sunlight, reducing efficiency during cloudy or rainy conditions.
- There are energy conversion losses (approximately 10–20%) due to electrical components and power regulation circuits.
- Battery capacity limitations restrict long-duration operation without sunlight.

- The mechanical tracking system may experience wear and tear over long-term use, requiring maintenance.

Despite these limitations, the system proves to be a cost-effective, efficient, and scalable renewable energy solution suitable for rural electrification and small-scale off-grid applications.

Chapter 5: Summary and Future Work

5. Summary

This project successfully develops a smart solar tracking system integrated with energy storage and multi-output applications. The use of Arduino-based control improves efficiency, while the TP4056 module ensures safe battery charging.

The system effectively:

- Maximizes solar energy capture
- Stores energy efficiently
- Supports practical applications like mobile charging and irrigation

Overall, it serves as a low-cost, eco-friendly, and standalone renewable energy solution suitable for real-world implementation.

6. Future Work

The system can be further improved in the following ways:

6.1. Dual-Axis Tracking

Implementing dual-axis tracking can further increase efficiency by tracking both horizontal and vertical movement of the sun.

6.2. IoT Integration

- Remote monitoring using mobile apps

- Real-time data (voltage, current, battery level)

3. MPPT (Maximum Power Point Tracking)

Replacing TP4056 with MPPT controller can:

- Increase charging efficiency
- Extract maximum power from solar panel

4. Higher Capacity Battery System

- Increase storage capacity
- Support larger loads (bigger pumps, appliances)

5. Smart Irrigation System

- Add soil moisture sensors
- Automate water pumping based on soil condition

6. Improved Mechanical Design

- More durable structure
- Weather-resistant materials